

A woman with long dark hair, wearing a red knit sweater, is looking down at a smartphone she is holding in her hands. A large, hand-drawn red speech bubble graphic is positioned over the phone and the word 'Vodafone'. The background is blurred, showing warm, bokeh-style lights.

Vodafone

Omnia Ibrahim | Data Scientist



Vodafone is one of the world's largest telecoms operators

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Introduction



- **Overview of the project:**
The data highlights employee activity in terms of value, cost, and approval status, helping assess performance, productivity, and resource efficiency.



Data Preprocessing

- *Collecting Employees' Data from Different excel sheets (sheet for each region) into only one sheet*
- *Handled missing values*
- *Corrected data inconsistencies*
- *Ensured data accuracy for performance and HR cost analysis.*
- *Merge Employee_Dept table with HR cost Table*
- *Extract relationship between the Employees table and HR cost table using EMP_ID*



Before Preprocessing

	A	B	C	D	E	F	G	H	I	J	K
1	DIM	Task	APPROVAL	Nov 21	Dec 21	Jan 22	Feb 22	Mar 22	Apr 22	May 22	Jun 22
2	EMP_0431										
3	Contracted Hours			165	157.5	150	150	172.5	150	150	
4	Allocated Hours			178.2	170.1	162	222	186.3	162	162	1
5	Remaining			-13.2	-12.6	-12	-72	-13.8	-12	-12	
6	Technical	TASK_T0958	Completed	13.2	12.6	12	12	13.8	12	12	
7	Technical	TASK_T0690	Completed	165	157.5	150	150	0	0	0	
8	Technical	TASK_T0740	Completed	0	0	0	60	30	0	0	
9	Technical	TASK_T0010	Completed	0	0	0	0	142.5	150	150	
10	EMP_0265										
11	Contracted Hours			165	157.5	150	150	172.5	150	150	
12	Allocated Hours			181.5	168	149.25	189	224.25	195	183.75	5
13	Remaining			-16.5	-10.5	0.75	-39	-51.75	-45	-33.75	10
14	Technical	TASK_T0437	Completed	0	0	0	24	34.5	30	30	
15	Technical	TASK_T0477	Completed	0	0	0	7.5	8.63	7.5	7.5	
16	Technical	TASK_T0476	Completed	0	0	0	15	17.25	15	15	
17	Technical	TASK_T0729	Completed	14.85	14.17	6.75	0	0	0	0	
18	Technical	TASK_T0853	Completed	1.65	0	0	0	0	0	0	
19	Technical	TASK_T0844	Completed	112.2	110.25	105	105	120.75	105	105	
20	Technical	TASK_T0398	Completed	11.55	4.2	0	0	0	0	0	
21	Technical	TASK_T0426	Completed	41.25	39.38	37.5	37.5	43.13	37.5	26.25	
22	EMP_0001										

Central USA Lat_Pa AF MEA MER CSE Sub-Saharan F_east Employee Dept Employee HR Cost

Ready Accessibility: Investigate

130%

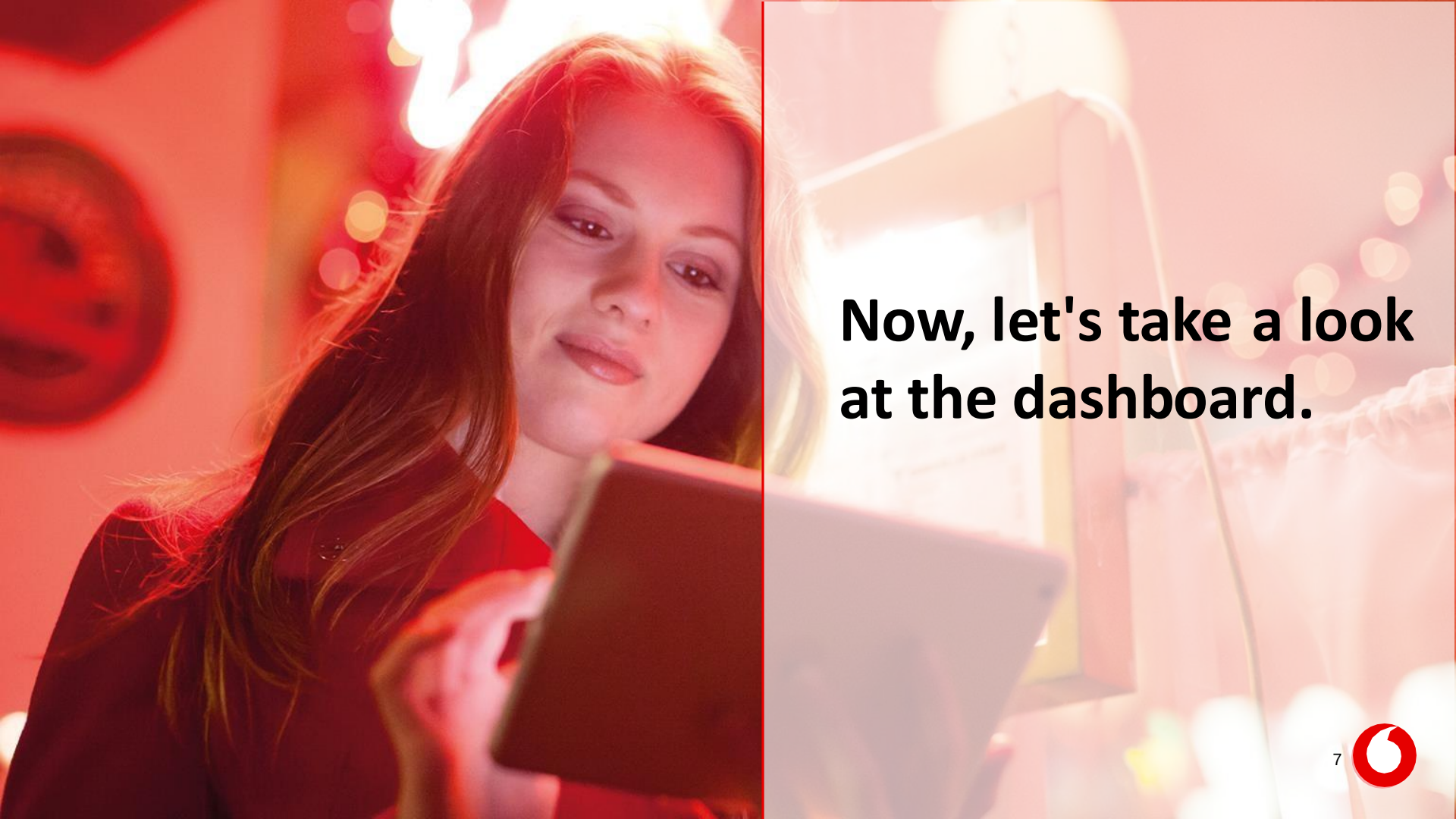


After Preprocessing

	A	B	C	D	E	F	G	H	I	J	K
1	Location	Dim	Sub-Dim	Task	APPROVAL	Date	Value	LEVEL-01	LEVEL-02	LEVEL-03	Cost
2	AF	EMP_0431	Contracted Hours	N/A	N/A	21-Nov	165	DEPT_02	Grade_1.2	LVL_01	7.887198
3	AF	EMP_0431	Allocated Hours	N/A	N/A	21-Nov	178.2	DEPT_02	Grade_1.2	LVL_01	7.887198
4	AF	EMP_0431	Remaining	N/A	N/A	21-Nov	-13.2	DEPT_02	Grade_1.2	LVL_01	7.887198
5	AF	EMP_0431	Technical	TASK_T0690	Completed	21-Nov	165	DEPT_02	Grade_1.2	LVL_01	7.887198
6	AF	EMP_0431	Technical	TASK_T0010	Completed	21-Nov	0	DEPT_02	Grade_1.2	LVL_01	7.887198
7	AF	EMP_0431	Technical	TASK_T0958	Completed	21-Nov	13.2	DEPT_02	Grade_1.2	LVL_01	7.887198
8	AF	EMP_0431	Technical	TASK_T0740	Completed	21-Nov	0	DEPT_02	Grade_1.2	LVL_01	7.887198
9	Central USA	EMP_0431	Contracted Hours	N/A	N/A	21-Nov	165	DEPT_02	Grade_1.2	LVL_01	7.887198
10	Central USA	EMP_0431	Allocated Hours	N/A	N/A	21-Nov	178.2	DEPT_02	Grade_1.2	LVL_01	7.887198
11	Central USA	EMP_0431	Remaining	N/A	N/A	21-Nov	-13.2	DEPT_02	Grade_1.2	LVL_01	7.887198
12	MER	EMP_0431	Contracted Hours	N/A	N/A	21-Nov	165	DEPT_02	Grade_1.2	LVL_01	7.887198
13	MER	EMP_0431	Allocated Hours	N/A	N/A	21-Nov	178.2	DEPT_02	Grade_1.2	LVL_01	7.887198
14	MER	EMP_0431	Remaining	N/A	N/A	21-Nov	-13.2	DEPT_02	Grade_1.2	LVL_01	7.887198
15	MER	EMP_0431	Technical	TASK_T0958	Completed	21-Nov	13.2	DEPT_02	Grade_1.2	LVL_01	7.887198
16	MER	EMP_0431	Technical	TASK_T0690	Completed	21-Nov	165	DEPT_02	Grade_1.2	LVL_01	7.887198
17	MER	EMP_0431	Technical	TASK_T0740	Completed	21-Nov	0	DEPT_02	Grade_1.2	LVL_01	7.887198
18	MER	EMP_0431	Technical	TASK_T0010	Completed	21-Nov	0	DEPT_02	Grade_1.2	LVL_01	7.887198
19	AF	EMP_0431	Contracted Hours	N/A	N/A	21-Dec	157.5	DEPT_02	Grade_1.2	LVL_01	7.887198

employee_agg_final_final





**Now, let's take a look
at the dashboard.**

Performance Overview Dashboard

DIM

1

#

3.65K

Location

3

Tasks

1

Approval Stauses

1

Date

01/11/2021

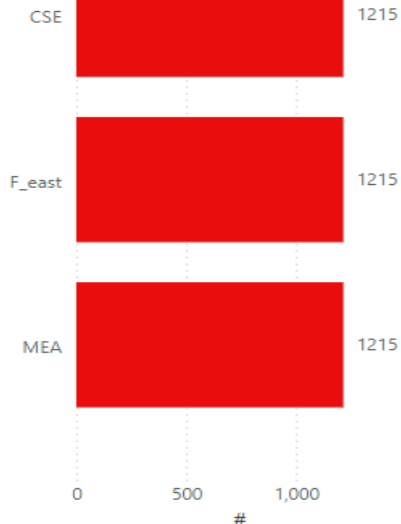


01/02/2023

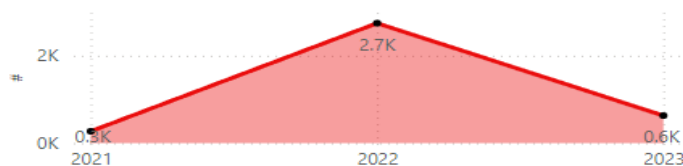


Trend

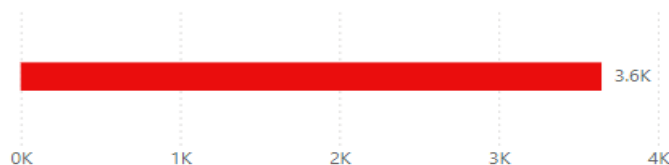
Locations Analysis



Trend Analysis



Task Analysis



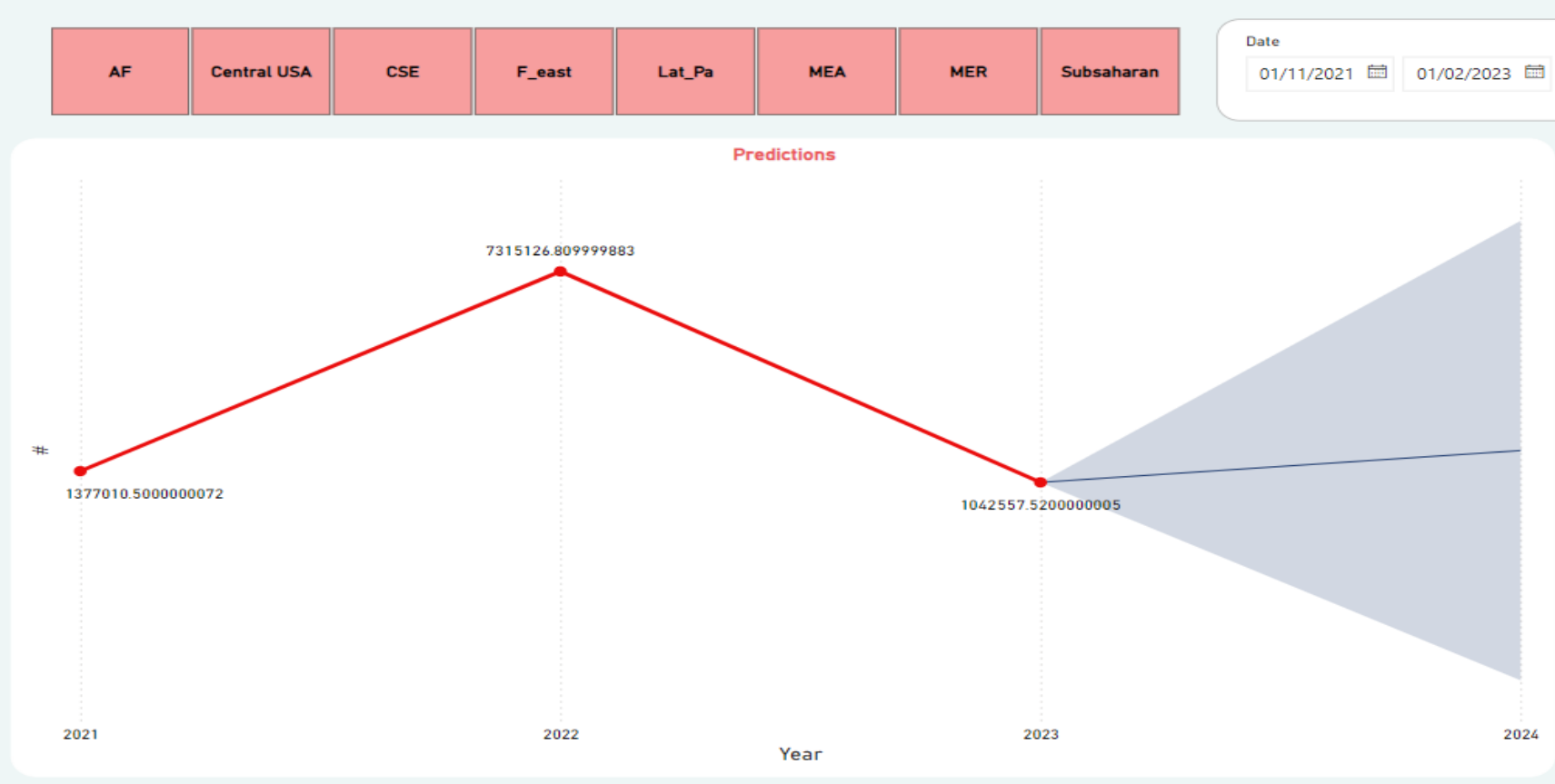
Sub DIM Analysis



Location	DIM	Sub DIM	Task	APPROVAL	Year	Quarter
CSE	EMP_0002	Allocated Hours			2021	Qtr 4
CSE	EMP_0002	Allocated Hours			2021	Qtr 4
CSE	EMP_0002	Allocated Hours			2022	Qtr 1
CSE	EMP_0002	Allocated Hours			2022	Qtr 1
CSE	EMP_0002	Allocated Hours			2022	Qtr 1
CSE	EMP_0002	Allocated Hours			2022	Qtr 2
CSE	EMP_0002	Allocated Hours			2022	Qtr 2
CSE	EMP_0002	Allocated Hours			2022	Qtr 2
CSE	EMP_0002	Allocated Hours			2022	Qtr 3
CSE	EMP_0002	Allocated Hours			2022	Qtr 3
CSE	EMP_0002	Allocated Hours			2022	Qtr 3
CSE	EMP_0002	Allocated Hours			2022	Qtr 4
CSE	EMP_0002	Allocated Hours			2022	Qtr 4
CSE	EMP_0002	Allocated Hours			2022	Qtr 4



Predictions Insights Dashboard



Key Influencers Dashboard

Key influencers Top segments

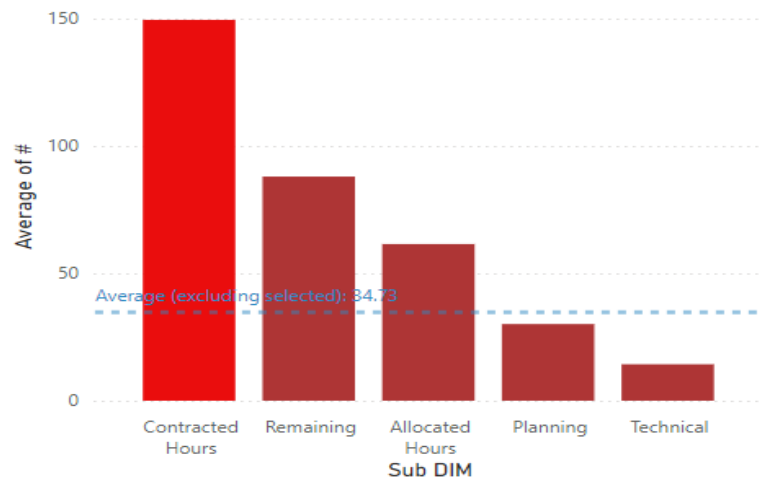
What influences # to ?

When...

....the average of # increases by



← # is more likely to increase when Sub DIM is Contracted Hours than otherwise (on average).



☐ Only show values that are influencers



Employee Tracker Dashboard

Employees
788

Cost
9.59

Level 1
3

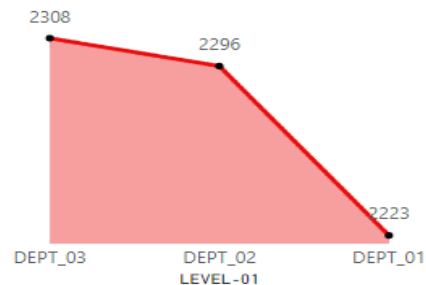
Level 2
3

Level 3
4

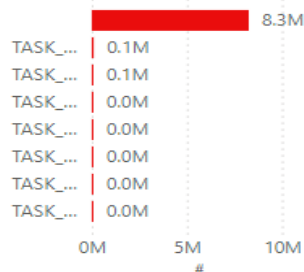
Average Cost
9.59

Tasks
969

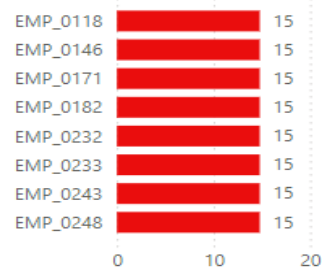
Level 1 Analysis



Task Analysis

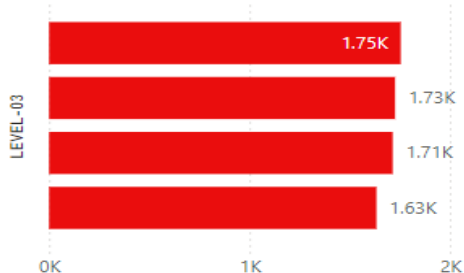


Locations Analysis

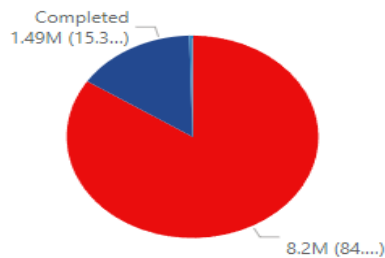


Employee	LEVEL-01	LEVEL-02	LEVEL-03	Cost
EMP_0033	DEPT_01	Grade_1.2	LVL_01	10.12
EMP_0076	DEPT_01	Grade_1.2	LVL_01	10.12
EMP_0081	DEPT_01	Grade_1.2	LVL_01	10.12
EMP_0083	DEPT_01	Grade_1.2	LVL_01	10.12
EMP_0134	DEPT_01	Grade_1.2	LVL_01	10.12
EMP_0158	DEPT_01	Grade_1.2	LVL_01	10.12
EMP_0159	DEPT_01	Grade_1.2	LVL_01	10.12
EMP_0267	DEPT_01	Grade_1.2	LVL_01	10.12
EMP_0311	DEPT_01	Grade_1.2	LVL_01	10.12
EMP_0326	DEPT_01	Grade_1.2	LVL_01	10.12

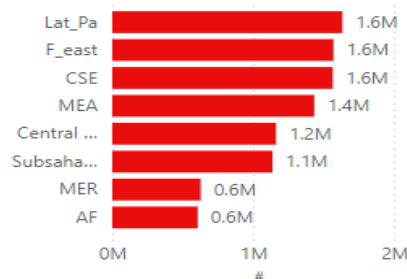
Level 3



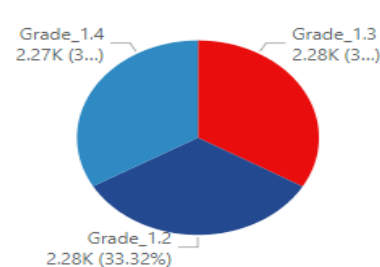
Approval Status Analysis



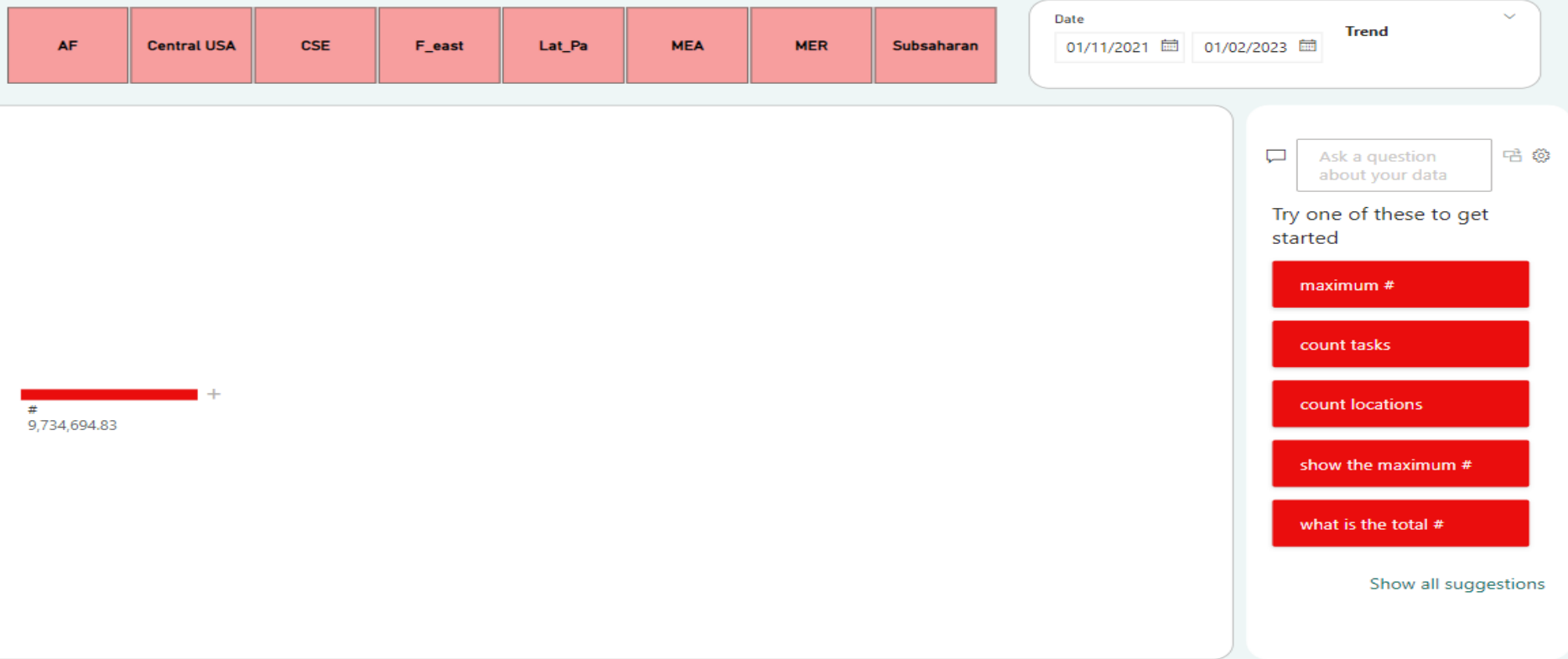
Locations Analysis



Level 2



Draw Your Report Dashboard



Conclusion

- The data gives an overview of employee activity by showing key metrics like value, cost, and approval status. It helps in understanding performance and productivity by comparing contracted, allocated, and remaining hours, offering insights into how efficiently resources are used and whether goals are being met. The analysis supports better task distribution and resource management to boost productivity and performance.





The future is exciting.

Ready?

